Bipartition Covers in the Multispecies Coalescent Model

Zachary McNulty

UC Berkeley

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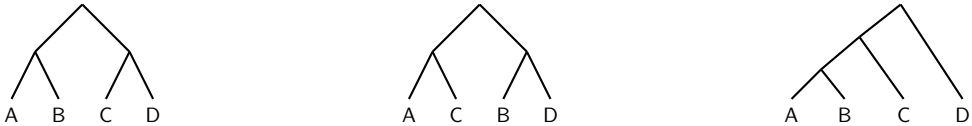
Talk Outline

- Background: phylogenetic inference and MSC model
- Motivation: superexponential tree search and bipartition covers
- Identify worst-case topologies and a balancing lemma.
- Empirical results
- Asymptotic analysis

Background: Phylogenetic Inference

Goal: Given a collection of k species, estimate their evolutionary history (*species tree*)

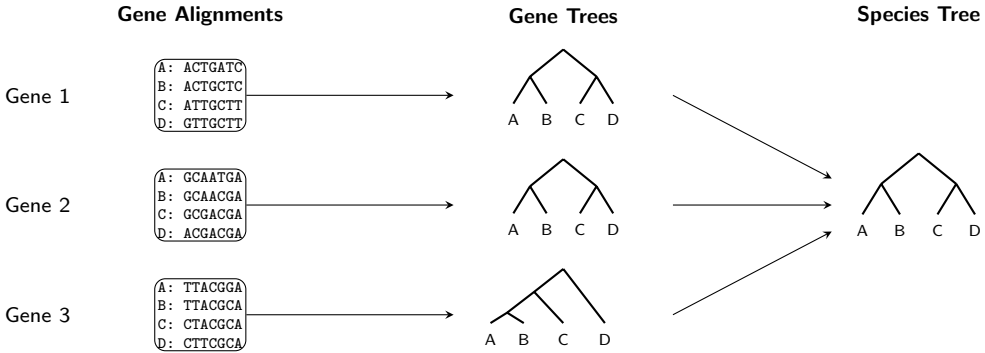


Challenge: We only have modern day DNA sequences, not the actual evolutionary history.

Background: Phylogenetic Inference

In *consensus methods*, this occurs in two stages:

- For each gene, generate an estimate of its evolutionary history (*gene tree*).
- Aggregate these into a single estimate of the species tree.



Background: Kingman's Coalescent

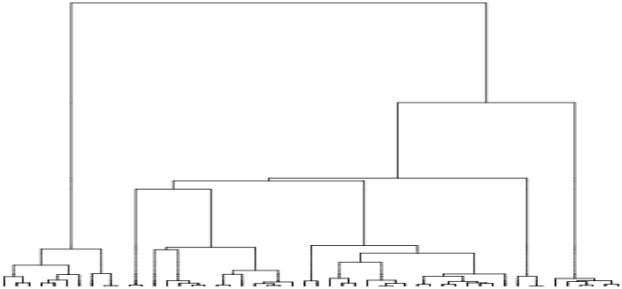
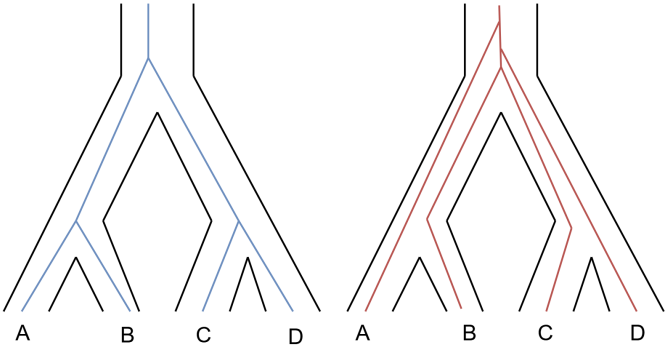


Figure: Kingman's coalescent process: Adwiladan, CC BY-SA 4.0, via Wikimedia Commons.

Start with n lineages. Each pair of lineages coalesces at rate one.

Background: Multispecies Coalescent Model



The *multispecies coalescent model (MSC)* models gene trees directly.

- Along each branch, each pair of lineages coalesce at rate one (Kingman's coalescent)
- Gene tree topologies can differ from species tree.

Motivation: Search-Space Bottleneck

Many consensus methods are framed as optimization problems over the space of trees:

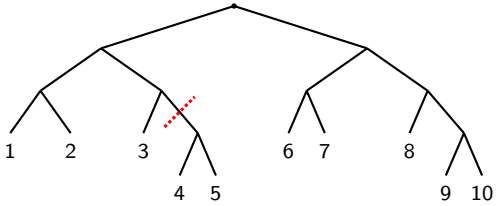
$$\hat{\mathcal{T}} = \arg\max_{\mathcal{T}} \mathrm{support}(\mathcal{T}, \mathcal{G})$$

Issue: There are $(2k-3)!!$ possible rooted binary phylogenetic trees.

- At $k = 20$ this is $\approx 8.2 \times 10^{21}$.
- We must find some way to restrict search space.

Motivation: Bipartition Covers

Every edge e in a tree $\mathcal{T}$ generates a *bipartition* $A \mid B$ of its leaves.



The bipartition associated to the highlighted edge is $\{4, 5\} \cup \{1, 2, 3, 6, 7, 8, 9, 10\}$.

Denote $\Sigma(\mathcal{T})$ the set of bipartitions. We say a set $\mathcal{G}$ of trees forms a *bipartition cover* if:

$$\Sigma(\mathcal{T}) \subseteq \bigcup_{g \in \mathcal{G}} \Sigma(g)$$

Problem Setting

Idea: consider only species trees whose bipartitions are present in the gene trees.

Namely, we restrict our search space to:

$$\mathcal{A} := \left\{ \mathcal{T} \ : \ \Sigma(\mathcal{T}) \subseteq \bigcup_{g \in \mathcal{G}} \Sigma(g) \right\}$$

Our Question: How many genes are required to ensure $\mathcal{T}^* \in \mathcal{A}$ with high probability?

Project goals

In this project, we aim to develop:

- Topology-free bounds on coverage probability $\mathbb{P}(\mathcal{T} \in \mathcal{A})$.
- Deeper understanding of "worst-case" topologies under MSC.
- Asympototic analysis of absorption times in Kingman's coalescent.

Key Coalescent Probability

For $1 \leq j \leq i$, define:

$$g_{i,j}(t) = \mathbb{P}(i \text{ lineages coalesce into } j \text{ after time } t \text{ under Kingman's coalescent})$$
$$= \sum_{k=j}^{i} \frac{e^{-\binom{k}{2}t}(2k-1)(-1)^{k-j} j_{(k-1)} i_{[k]}}{j!(k-j)! i_{(k)}}$$

In particular, $g_{\ell,1}(t)$ is the probability for full coalescence of ℓ lineages by time t.

Properties $g_{\ell,1}(t)$:

- Increasing, log-concave in t. Decreasing, convex in ℓ.
- Can view as the CDF of $\sum_{i=2}^{\ell} \mathrm{Exp}\left(\binom{i}{2}\right)$.
- Bound away from zero: $s(t) := \lim_{\ell \to \infty} g_{\ell,1}(t) > 0$.

Union Bound

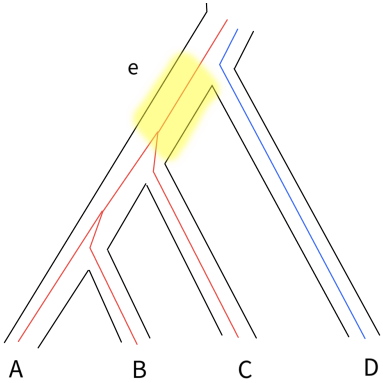
A rooted binary tree on k leaves has $k-3$ nontrivial bipartitions.

Let $E_{n,\ell}$ be the event the split associated to edge e_ℓ is found at least once in n gene trees.

$$\mathbb{P}(\text{cover}) = \mathbb{P}\Big(\bigcap_{\ell=1}^{k-3} E_{n,\ell}\Big) \geq 1 - \sum_{\ell=1}^{k-3} \mathbb{P}(E_{n,\ell}^c) = 1 - \sum_{\ell=1}^{k-3} (1 - \mathbb{P}(E_{1,\ell}))^n,$$

Let F_e be the event that all lineages below the given edge e coalesce by the time they exit e, and $p_e = \mathbb{P}(F_e)$. Clearly then $\mathbb{P}(E_{1,\ell}) \geq p_{e_\ell}$.

The event F_e



F_e: all lineages below the highlighted edge e coalesce before they exit e.

Bottlenecks

Heuristically, p_e depends on:

- The number of lineages k_e below edge e.
- The subtree rooted below e.

To develop topology-free bounds, we must overcome two bottlenecks:

- **Combinatorial:** tree topologies with many large bipartitions.
- **Coalescent:** tree topologies which make coalescence unlikely.

Extremal Topologies

Caterpillar tree

Balanced tree

Figure: The two main extremal tree topologies: caterpillar and balanced trees.

Caterpillar Bound

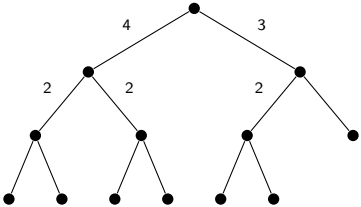


Figure: Example of the descendant counts $\{\alpha_i\}$ associated to the nontrivial bipartitions/edges.

Lemma

If $1 < \alpha_1 \leq \ldots \leq \alpha_{k-3}$ are the descendant counts of any phylogenetic tree, then $\alpha_i \leq i + 1$. In particular, if $f : \mathbb{N} \to \mathbb{R}$ is increasing then:

$$\sum_{i=1}^{k-3} f(\alpha_i) \leq \sum_{i=2}^{k-2} f(i)$$

Balanced Trees Minimize Coalesence

FOSD: Given X, Y we say $X \leq_{st} Y$ if $\mathbb{E}\phi(X) \leq \mathbb{E}\phi(Y)$ for all increasing functions ϕ.

Let $X_{\mathcal{T}}$ be the number of lineages reaching the root of a tree $\mathcal{T}$ under the MSC model.

Theorem

If $\mathcal{T}$ has k leaves and minimum branch length t_{min}, then:

$$X_{\mathcal{T}} \leq_{st} X_{\mathcal{B}_{k,t_{min}}}$$

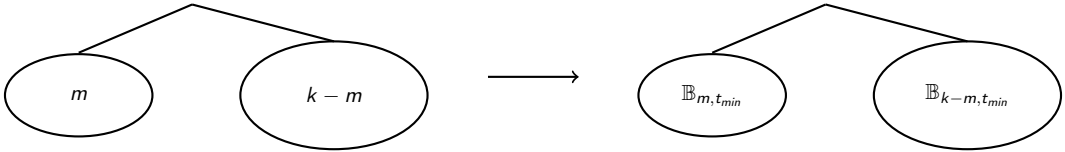
where $\mathcal{B}_{k,t}$ is the balanced tree with k leaves and all branch lengths equal to t_{min}.

As $g_{\ell,1}(t)$ is a decreasing function of ℓ, we see:

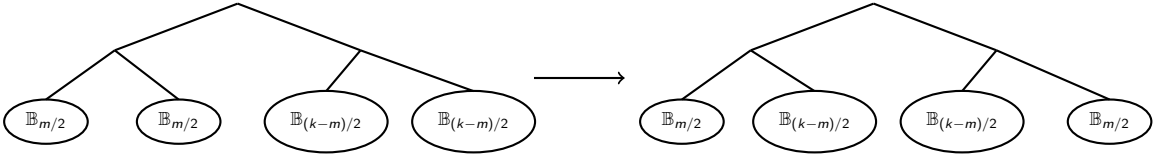
$$p_e := \mathbb{E}g_{X_e,1}(t_e) \geq \mathbb{E}g_{W_e,1}(t_{min}), \quad W_e := X_{\mathcal{B}_{k_e,t_{min}}}.$$

Proof Outline

By induction on k, we see $X_{\mathcal{T}} \leq_{st} X_{\mathcal{T}'}$ where $\mathcal{T}'$ replaces subtrees of root with balanced trees:



Next, we want to rebalance the subtrees so they have the same size:



Deterministic Balancing Lemma

Let Z_i^t have distribution $\mathbb{P}(Z_i^t = j) = g_{ij}(t)$.

Lemma (Deterministic Balancing Lemma)

Fix $k \in \mathbb{N}$. For $1 \leq i \leq j \leq \lfloor \frac{k}{2} \rfloor$:

$$Z_i^t + Z_{k-i}^t \leq_{st} Z_j^t + Z_{k-j}^t$$

Proof Outline

Deterministic Balancing Lemma:

- Condition on time of first coalescent event; more balanced = longer wait.

$$\text{rate} = \binom{i}{2} + \binom{k-i}{2}$$

- Rely on monotonicity with respect to t: for $t < t'$, $Z_i^t \geq_{st} Z_i^{t'}$.
- After first coalescent event, imbalance can only change by one. Hence if $i < j < \lfloor k/2 \rfloor$ starts more imbalanced, it must tend to stay that way.

$$(i, k-i) \text{ with imbalance } k-2i \to \begin{cases} (i-1, k-i) & \text{with imbalance } k-2i+1 \\ (i, k-i-1) & \text{with imbalance } k-2i-1 \end{cases}$$

Balancing Lemma

Now extend to a random number of initial lineages.

Likelihood Ratio Order: We say $X \leq_{lr} Y$ if:

$$\frac{\mathbb{P}(X = k)}{\mathbb{P}(Y = k)}$$

is non-decreasing in k. Note $X \leq_{lr} Y$ implies $X \leq_{st} Y$.

Lemma (Balancing Lemma)

If $A \leq_{lr} A'$ and $B \leq_{lr} B'$, and A, B, A', B' are mutually independent then:

$$Z^t_{A+B} + Z^t_{A'+B'} \leq_{st} Z^t_{A+B'} + Z^t_{A'+B}$$

Proof Outlines

Balancing Lemma:

- Rephrase deterministic lemma in terms of sums/differences: if $0 \leq d \leq d' < s$ then:

$$Z^t_{\frac{s-d'}{2}} + Z^t_{\frac{s+d'}{2}} \leq_{st} Z^t_{\frac{s-d}{2}} + Z^t_{\frac{s+d}{2}}$$

- Extend to random differences: if $0 \leq_{st} D \leq_{st} D' \leq_{st} s$ then ...
- Define pairwise sums/differences

$$\Delta_A := A' - A, \quad \Delta_B := B' - B, \quad S_A := A' + A, \quad S_B = B' + B.$$

- Apply previous with conditional differences:

$$D' = |\Delta_A + \Delta_B| \mid \{|\Delta_A|, |\Delta_B|, S_A, S_B\}, \qquad D = |\Delta_A - \Delta_B| \mid \{|\Delta_A|, |\Delta_B|, S_A, S_B\}.$$

- After conditioning $\{D, D'\} = \{|\Delta_A| + |\Delta_B|, ||\Delta_A| - |\Delta_B||\}$.

Bipartition Cover Bound

Combining our above results gives:

Corollary (Bipartition Cover Bound)

Given n gene trees sampled independently under the MSC model from a species tree $\mathcal{T}$ with k species and minimum branch length t_{min}, then:

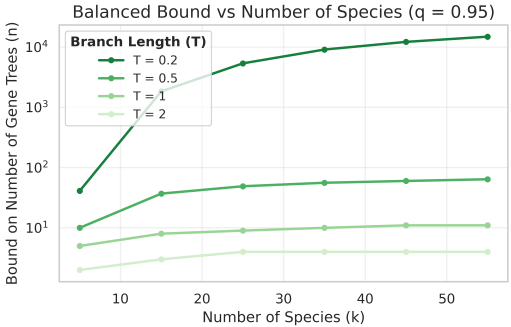
$$\mathbb{P}(cover) \geq 1 - \sum_{\ell=2}^{k-2} (1 - w_\ell)^n$$

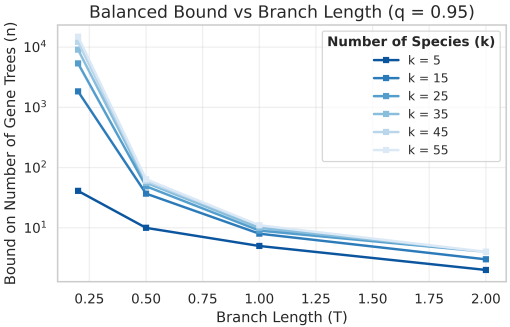
where $w_\ell := \mathbb{E} g_{W_\ell,1}(t_{min})$ for $W_\ell := X_{\mathcal{B}_{\ell,t_{min}}}$.

Due to recursive structure, distributions of W_ℓ are easy to compute dynamically.

Empirical Results

Fix $q \in (0,1)$. Let n_q be the smallest n that ensures $\mathbb{P}(\text{cover}) \geq q$ under our bound.





- Sharply sensitive to small t_{min}.
- Mostly saturates as a function of k past some point.

Overestimation Ratios

Empirically estimate true q^{th} quantile n_q^* for:

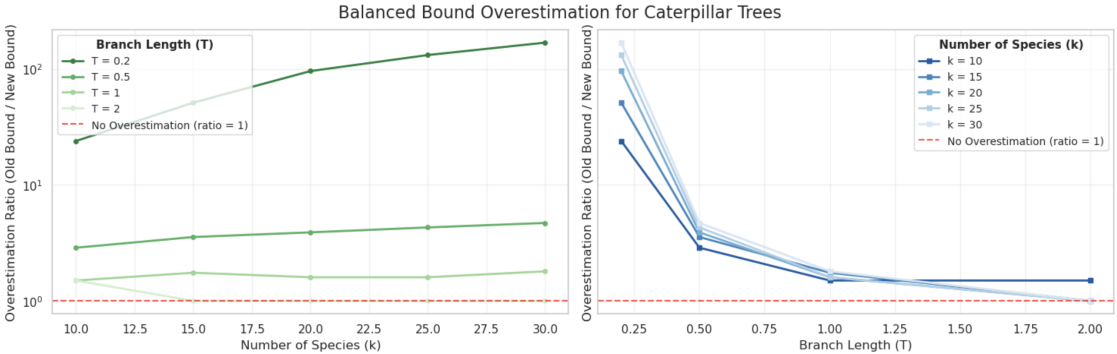
- Caterpillar trees with all branch lengths equal to t_{min}.
- Balanced trees with all branch lengths equal to t_{min}.

Then we plot the *overestimation ratio* $\frac{n_q}{n_q^*}$ as a function of k, t_{min}.

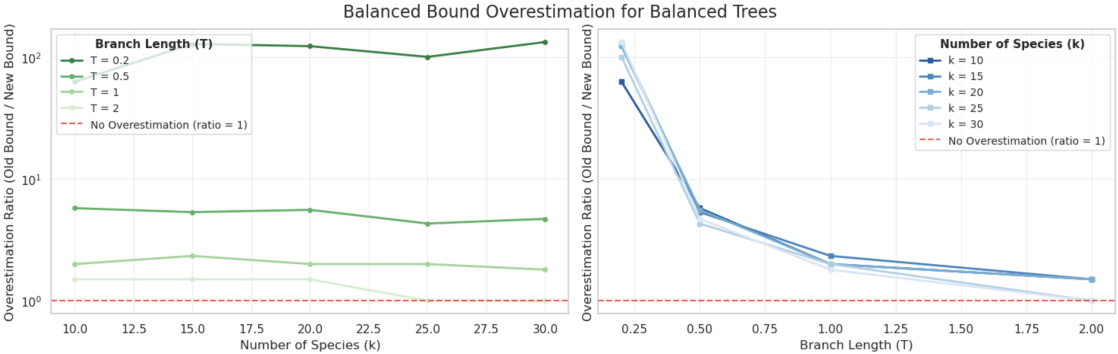
To get a sense of performance of our bound on a "typical" tree we:

- Simulate many random Yule trees.
- Estimate n_q^* for each tree via simulation.
- Study distribution of $\frac{n_q}{n_q^*}$ across trees.

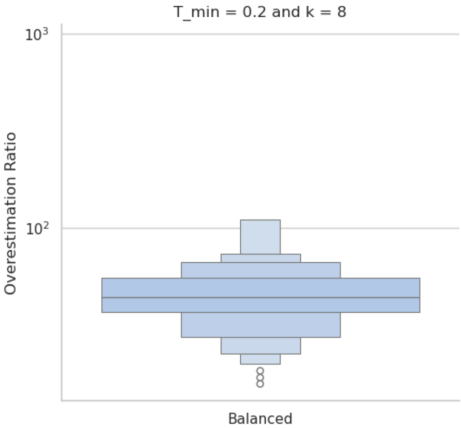
Overestimation Ratios: Caterpillar

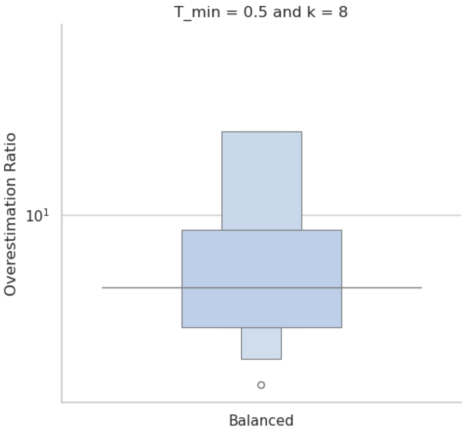


Overestimation Ratios: Balanced



Overestimation Ratios: Yule Trees





Asymptotic Analysis

We are interested in the regime where:

- The number of species k is large.
- The minimum branch length $t_{\min}$ is small.

Since n_q is only an implicit function of k, t_{min}, we upper bound it by observing:

- $g_{\ell,1}(t)$ is convex in ℓ.
- We can upper bound $\mathbb{E}W_\ell \leq \frac{2-e^{-t_{min}/2}}{1-e^{-t_{min}/2}} \asymp 2t_{min}^{-1}$. Reasonable for large k.

Hence:

$$\sum_{\ell=2}^{k-2}(1-w_\ell)^n \leq (k-3)(1-w_{k-2})^n \Rightarrow n_q \lesssim \frac{\log\left(\frac{1-q}{k-3}\right)}{\log(1-g_{2t^{-1},1}(t))}$$

Asymptotics of $g_{\ell,1}(t)$

In the fixed k, small t regime, we get polynomial decay in t^{-1}:

$$g_{\ell,1}(t) \sim \frac{\ell!}{2^{\ell-1}} t^{\ell-1}, \quad \ell^{-2} t \downarrow 0.$$

For fixed t we saw $g_{\ell,1}(t)$ saturates $s(t) := \lim_{\ell\to\infty} g_{\ell,1}(t) > 0$ with gap:

$$\Delta_t(k) := g_{\ell,1}(t) - s(t) = \frac{2s'(t)}{\ell} + o(\ell^{-1})$$

Moreover, via classical Tauberian theory the limiting CDF $s(t)$ has small t asymptotics:

$$s(t) = \exp\left(-\Theta(t^{-1})\right)$$

Large k, Small t_{min} Behavior

Suppose t_{min} is fixed. Let $h_t := g_{2t^{-1},1}(t)$ and $\Delta = h_t - s(t)$. Above we saw for large k:

$$n_q \lesssim \frac{\log(k)}{-\log(1-h_t)} \approx \frac{\log(k)}{s(t)\left(1+\frac{\Delta}{s(t)}\right)}$$

Some simple rearragements shows:

$$\frac{\Delta}{s(t)} \approx t \cdot \frac{d}{dt}\log s(t) \approx t^{-1}$$

so roughly:

$$n_q \lesssim \log(k)t\exp(\Theta(t^{-1}))$$

Takeaways

Takeaways:

- Caterpillars and balanced trees are the two main bottlenecks for bipartition recovery.
- Topology-free bounds are possible, but coarse; must account for *both* extremes.
- Due to saturation in k, the minimal branch length t_{min} is the key limiting factor.

Questions:

- Avoid union bound and take advantage of positive dependence?
- Extend to larger class of coalescent processes with sufficient "convexity"?

Key References

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